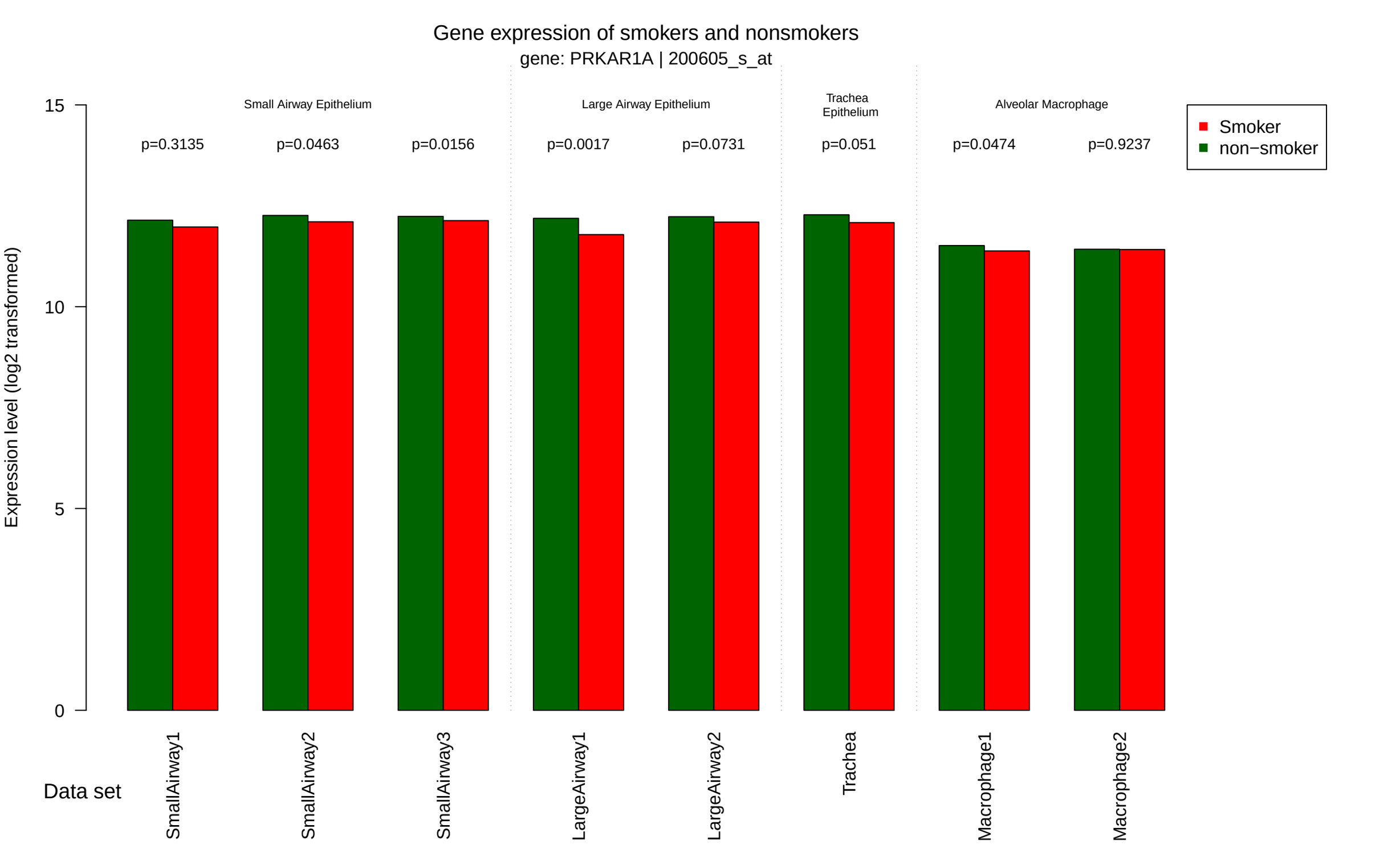
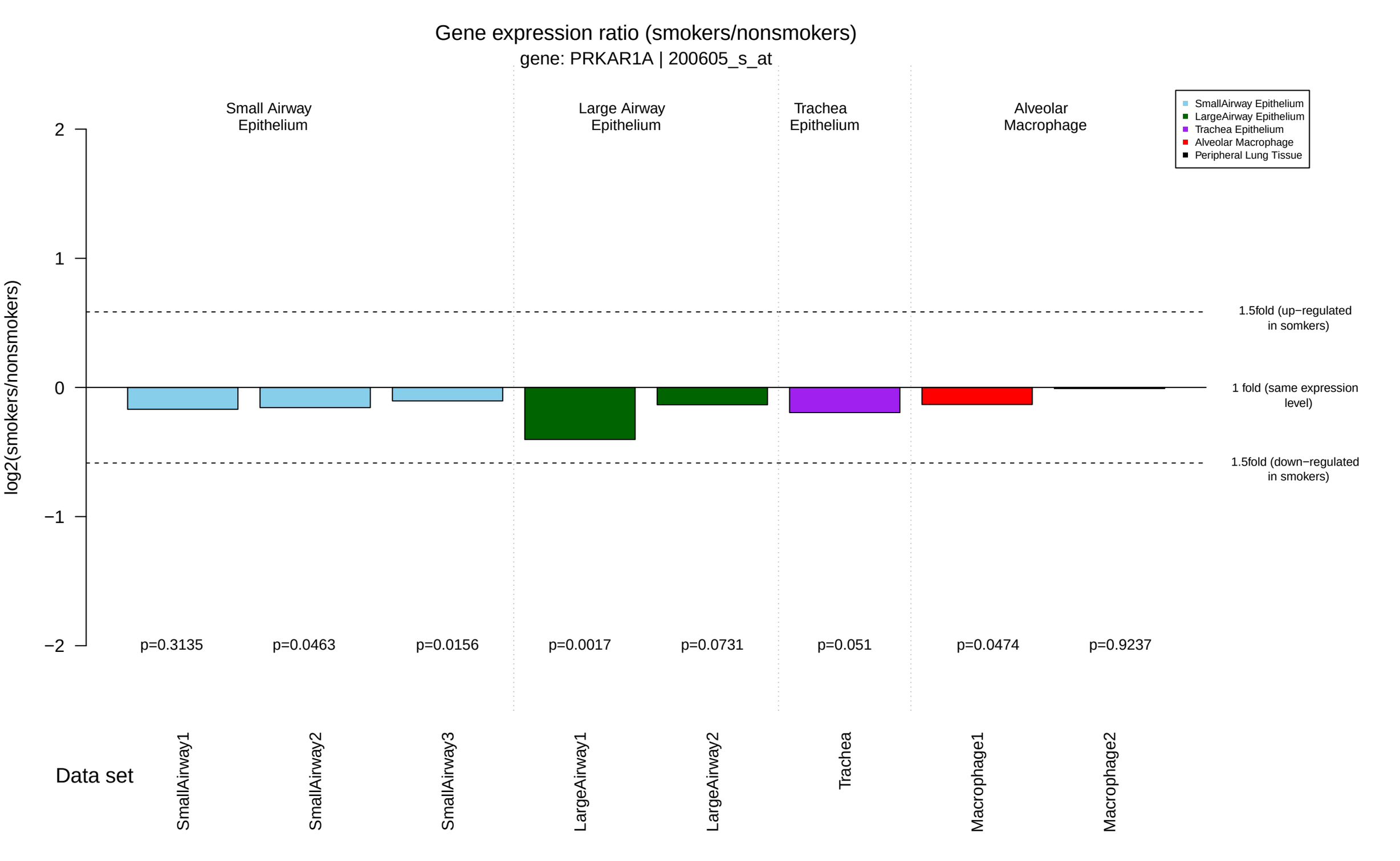
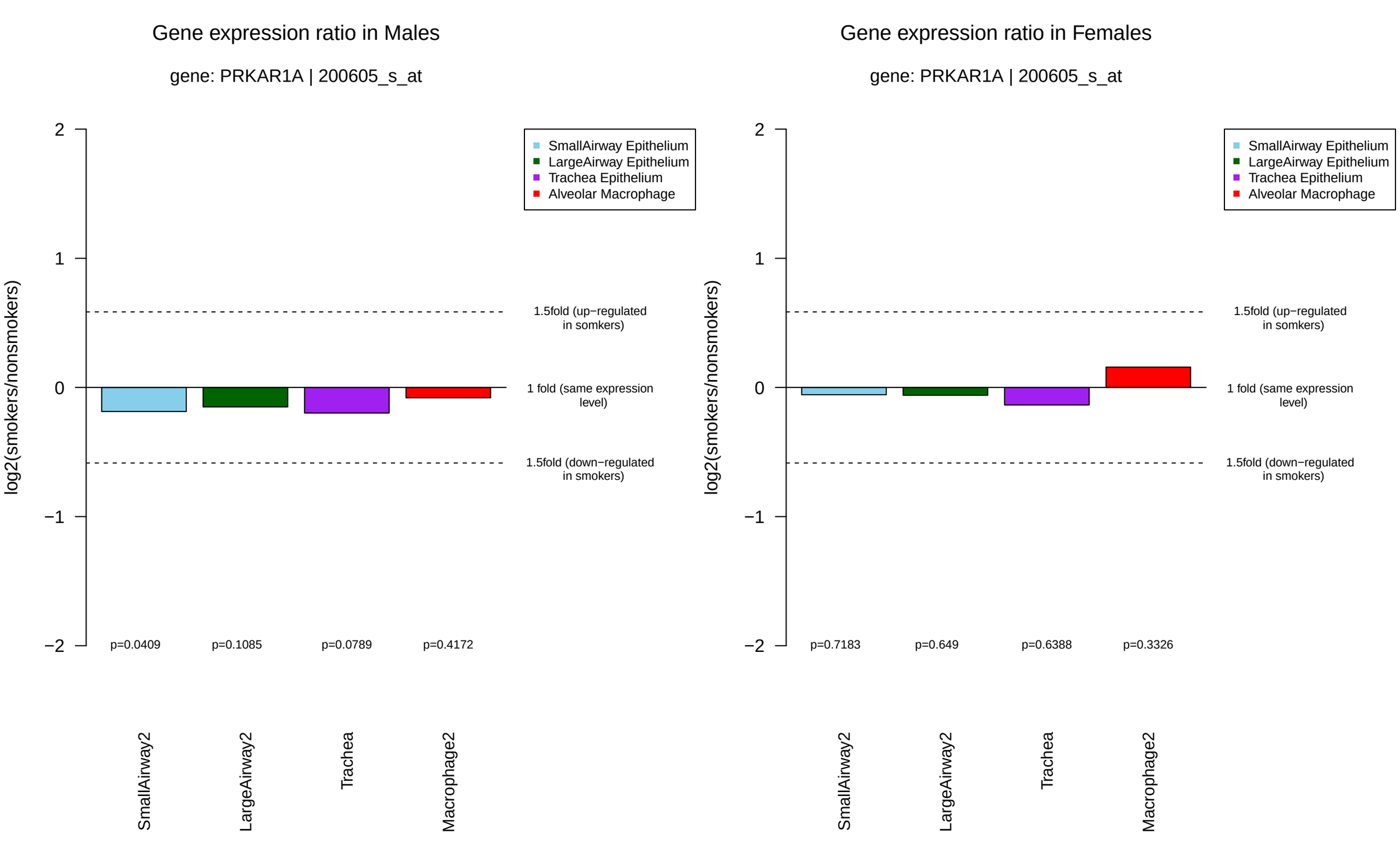
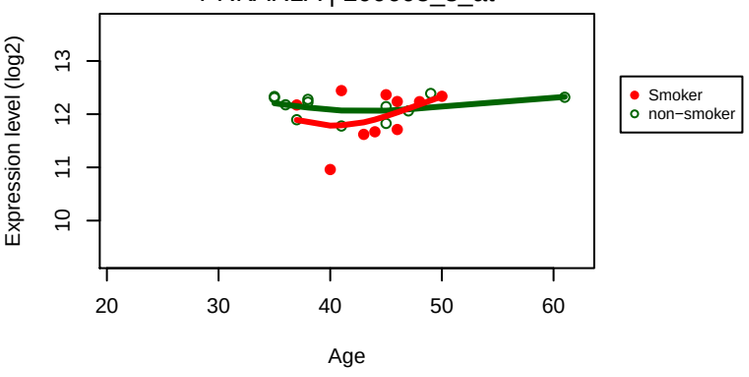
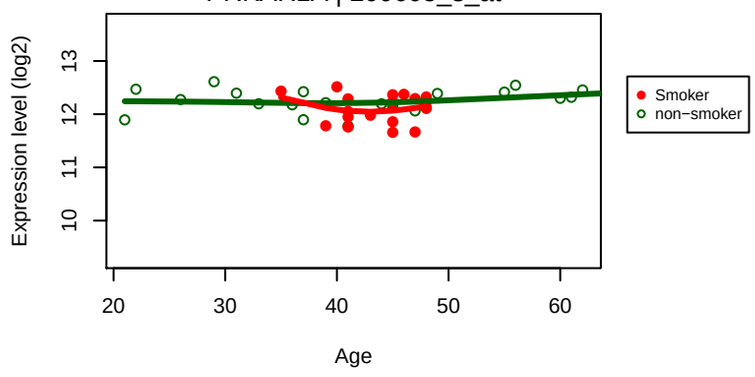
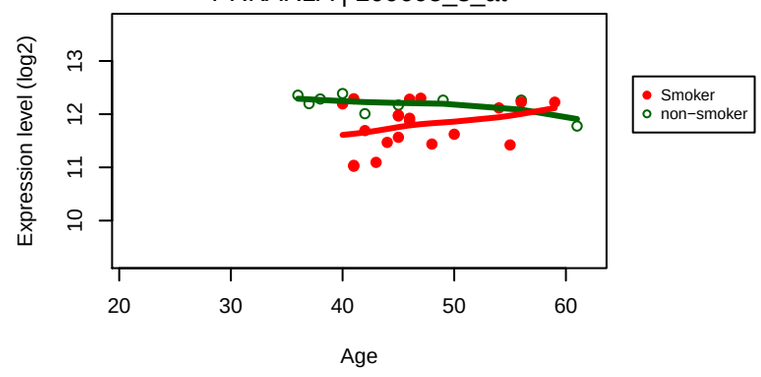

Gene expression of smokers and nonsmokers
gene: PRKAR1A | 200605_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3135
p=0.0463
p=0.0156
p=0.0017
p=0.0731
p=0.051
p=0.0474
p=0.9237
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

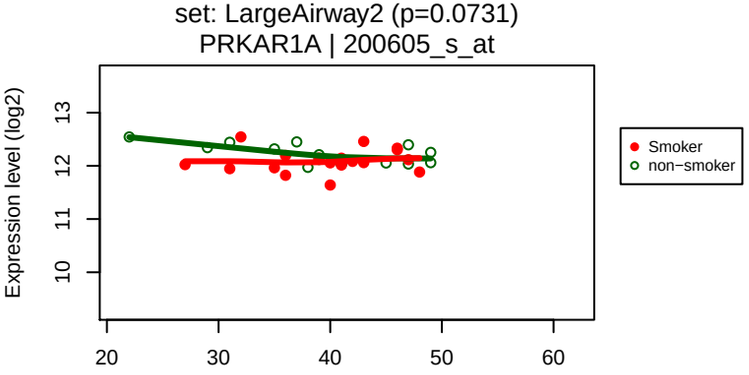

Gene expression ratio (smokers/nonsmokers)
gene: PRKAR1A | 200605_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3135
p=0.0463
p=0.0156
p=0.0017
p=0.0731
p=0.051
p=0.0474
p=0.9237
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

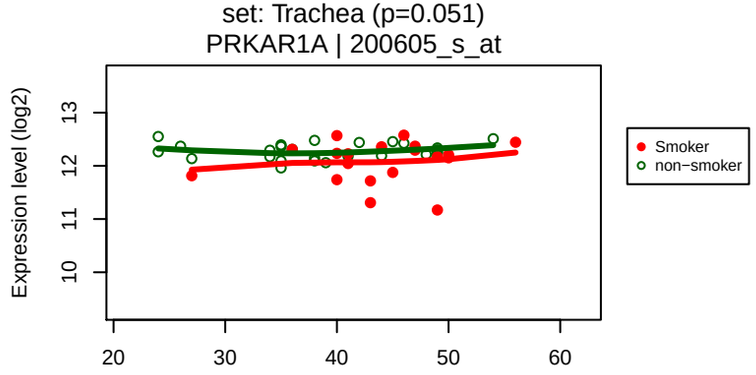

Gene expression ratio in Males
gene: PRKAR1A | 200605_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0409
p=0.1085
p=0.0789
p=0.4172
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PRKAR1A | 200605_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7183
p=0.649
p=0.6388
p=0.3326
SmallAirway2
LargeAirway2
Trachea
Macrophage2

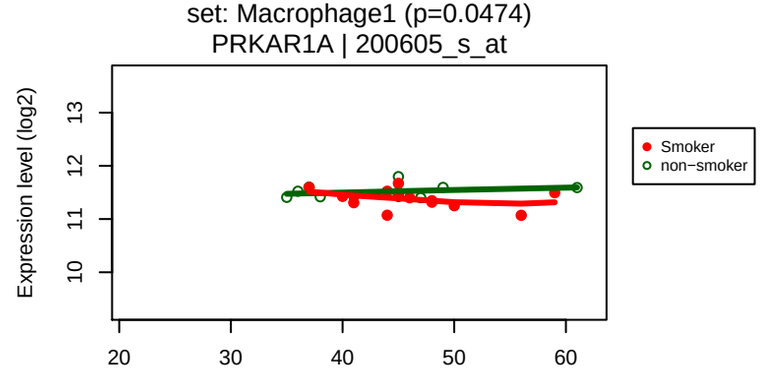

set: SmallAirway1 (p=0.3135)
PRKAR1A | 200605_s_at
Expression level (log2)
10
11
12
13
20
30
40
50
60
Age
Smoker
non−smoker

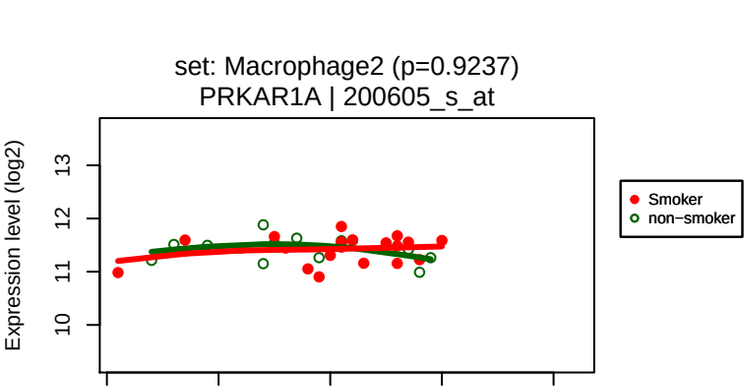

set: SmallAirway2 (p=0.0463)
PRKAR1A | 200605_s_at
Expression level (log2)
10
11
12
13
20
30
40
50
60
Age
Smoker
non−smoker

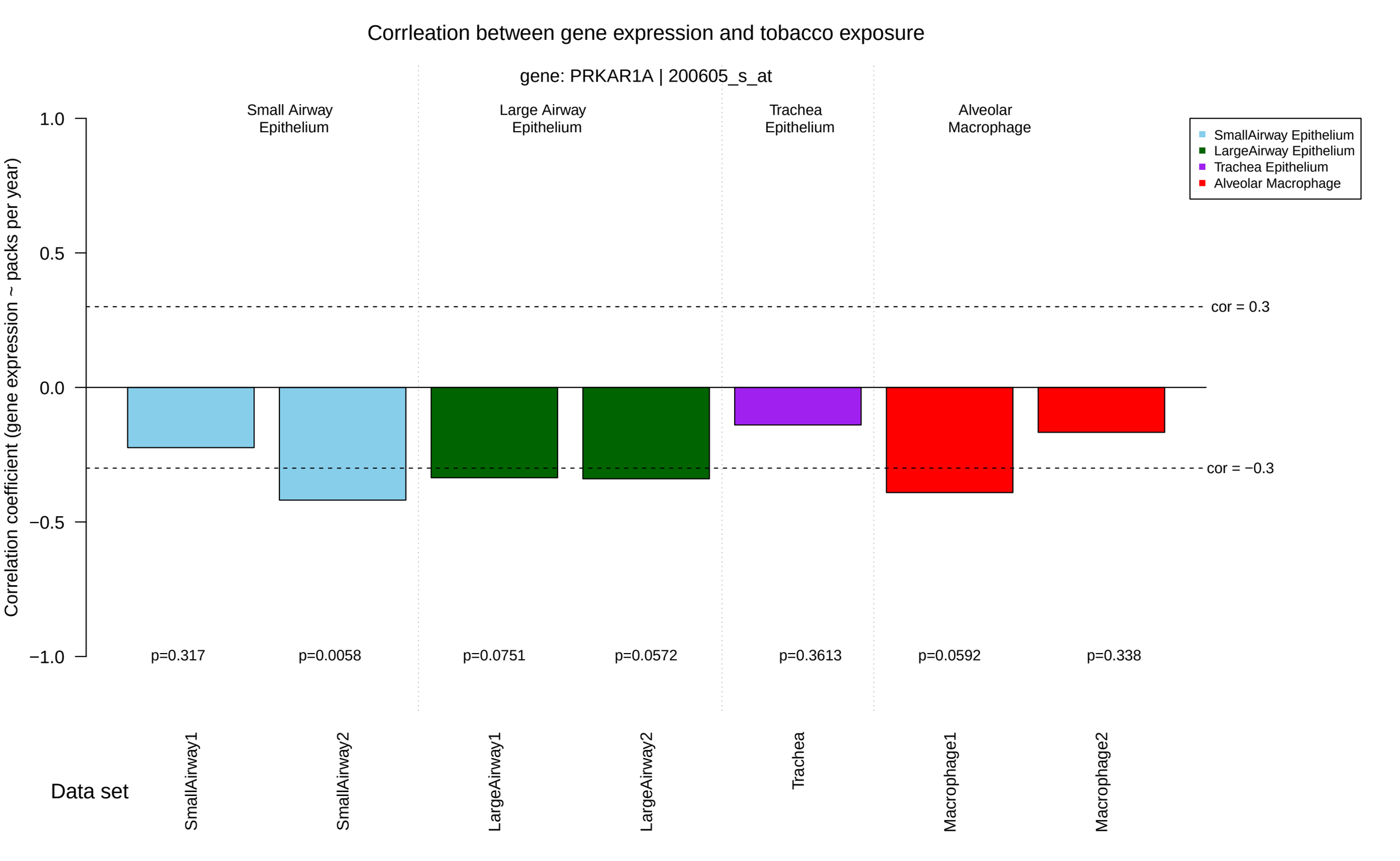

set: LargeAirway1 (p=0.0017)
PRKAR1A | 200605_s_at
Expression level (log2)
10
11
12
13
20
30
40
50
60
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0731)
PRKAR1A | 200605_s_at
Expression level (log2)
10
11
12
13
20
30
40
50
60
Age
Smoker
non−smoker


set: Trachea (p=0.051)
PRKAR1A | 200605_s_at
Expression level (log2)
10
11
12
13
20
30
40
50
60
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0474)
PRKAR1A | 200605_s_at
Expression level (log2)
10
11
12
13
20
30
40
50
60
Age
Smoker
non−smoker


set: Macrophage2 (p=0.9237)
PRKAR1A | 200605_s_at
Expression level (log2)
10
11
12
13
20
30
40
50
60
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PRKAR1A | 200605_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.317
p=0.0058
p=0.0751
p=0.0572
p=0.3613
p=0.0592
p=0.338
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2